

Smart City Complaint Analysis Report

1. Problem Statement

Modern cities receive a large number of citizen complaints related to garbage collection, water supply, road damage, street light failures, traffic congestion, drainage problems, and public safety issues. Managing and resolving these complaints efficiently is challenging. The objective of this project is to analyse complaint data and identify complaint trends, service performance, and recurring civic issues to support better decision-making.

2. Dataset Description

The dataset contains citizen complaints submitted to city authorities. Important fields include complaint category, complaint date, resolution date, department, location, complaint status, and other service-related information. Data cleaning was performed to improve data quality before analysis.

3. Data Cleaning and Preparation

- Removed duplicate records.
 - Handled missing values.
 - Converted date columns into datetime format.
 - Created Resolution Delay feature.
 - Created Severity Score based on complaint priority.
 - Generated additional metrics for analysis.
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4. KPI Definitions

- Total Complaints
- Resolved Complaints
- Resolution Rate
- Average Resolution Time
- High Priority Complaints
- Citizen Satisfaction Score

- Complaint Hotspot Zones
- Department Performance Score

These KPIs help city officials monitor operational efficiency and service quality.

5. Analysis and Findings

- Certain complaint categories occurred more frequently than others.
 - Some city zones experienced repeated civic issues.
 - Resolution time varied across departments.
 - High-priority complaints often required longer resolution periods.
 - Complaint trends showed fluctuations across different months.
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6. Dashboard Screenshots

Insert the following screenshots:

- KPI Dashboard
 - Complaint Category Analysis
 - Monthly Complaint Trend
 - Zone-wise Complaint Distribution
 - Department Performance Analysis
 - Heatmap Analysis
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7. Business Questions and Answers

1. Which complaint categories occur most frequently?

The analysis showed that a few complaint categories contributed to the majority of citizen complaints, indicating areas requiring immediate attention.

2. Which areas experience recurring civic problems?

Certain zones consistently reported higher complaint volumes, suggesting recurring infrastructure and service-related issues.

3. Which departments take the longest to resolve complaints?

Department performance analysis revealed that some departments required significantly more time to resolve complaints compared to others.

4. What factors reduce citizen satisfaction?

Long resolution times, repeated complaints, and delays in handling high-priority issues negatively affected citizen satisfaction.

5. What actions should city authorities prioritize?

Authorities should focus on improving response times, allocating resources to high-complaint zones, and addressing recurring infrastructure issues.

8. Recommendations

- Increase resources in complaint hotspot areas.
 - Improve response time for critical complaints.
 - Monitor department performance regularly.
 - Prioritize recurring infrastructure problems.
 - Use data-driven strategies for service planning.
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9. Future Scope

- Real-time complaint monitoring dashboard.
 - Predictive analytics for complaint forecasting.
 - GIS-based hotspot analysis.
 - Automated citizen feedback systems.
 - AI-powered complaint classification and prioritization.
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10. Conclusion

The Smart City Complaint Analysis project successfully identified complaint trends, service bottlenecks, and recurring civic issues. The insights obtained from the analysis can help city authorities improve operational efficiency, reduce complaint resolution time, and enhance overall citizen satisfaction.